

GRB 090530

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ABSTRACT

We present the complete timing and time-resolved spectral analysis of the single-pulse burst GRB 090530 (*Fermi*/GBM trigger bn090530760), the fourth burst of a 106-burst single-pulse campaign run on a staged, gate-reviewed pipeline. The burst is long and slow: $T_{90} = 134.3^{+1.3}_{-0.7}$ s inside the approved $(-2.4, 169.2)$ s window, with the most extreme pulse asymmetry of the campaign so far (two-sigmoid $\varphi = 0.0169 \pm 0.0018$, $R^2 = 0.92$ — the rise is ~ 60 times faster than the decay) and a correspondingly large spectral lag: scanning the cross-correlation fit window gives $\tau = 3.50^{+0.26}_{-0.25} \pm 0.48_{\text{win}}$ s (25–50 vs. 100–300 keV, soft lagging hard). Of the three minimum-variability estimators only the Haar cross-check returns a bound (< 8.0 s); the wavelet method fails to find a minimum and the windowed method’s admission gate rejects the burst — both recorded as estimator failures, not measurements. Spectroscopy over six Bayesian blocks plus the integrated spectrum tells a two-phase story. The first three blocks violate the synchrotron line of death (Band $\alpha = +0.62$ to $+0.47$, all $> -2/3$) and the first block carries the campaign’s strongest thermal candidate to date: a cutoff power law plus blackbody with $kT = 22.4 \pm 1.1$ keV that beats every photospheric-free model by $\Delta\text{AIC} = 3.1$ and passes the Wien-peak edge gate. The decay is different physics: α drops below the line, and the double-smoothly-broken power law wins both the integrated spectrum ($E_b \simeq 11.8$ keV, $E_p \simeq 69.9$ keV; a statistical tie with Band at $\Delta\text{AIC} = 1.3$) and block 4 narrowly (nearest competitor at $\Delta\text{AIC} = 0.8$). In the last block the three top-ranked models rail a slope bound and are reported invalid; the best valid model is again thermal-bearing. Every panel of the 24×7 model grid is rendered under a no-model-dropped rule; all numbers carry their provenance sidecars, and the inter-detector calibration rail (b_0 at 0.80 throughout; n_2 at 1.20 in the integrated and late-bin winners) is disclosed as a standing caveat.

Keywords: Gamma-ray bursts (629) — Time domain astronomy (2109)

1. INTRODUCTION

GRB 090530 triggered the *Fermi* Gamma-ray Burst Monitor (Meegan et al. 2009) on 2009 May 30. As burst #4 of the single-pulse campaign it runs on the pipeline frozen from bursts #1–#2, and it extends the campaign’s central empirical axis — the relation of single-pulse spectra to the synchrotron line of death and to thermal components — in a direction neither earlier burst reached: both behaviours inside one burst. The first two campaign bursts bracketed the question at the burst level (one violator without thermal components, one compliant burst with a marginal thermal candidate); GRB 090530 shows the violation and the thermal component are *epoch-level* properties. Its integrated spectrum is also the campaign’s first won by the double-smoothly-broken power law, the two-break shape of Ravasio et al. (2019) that motivates the survey.

2. OBSERVATIONS, DETECTORS, AND APPROVED SELECTIONS

We use TTE data from NaI detectors n_1 , n_2 , n_5 and BGO b_0 (Figure 1), with the spectral convention of Chand et al. (2020): NaI 8–900 keV excluding the 30–40 keV K-edge, BGO 0.2–38 MeV. Background and source selections are the human-approved Stage-1 decisions (source window $(-2.43, 169.24)$ s; Figures 2–3); the same selections drive spectroscopy and timing. Inter-detector effective-area factors float in $[0.8, 1.2]$; b_0 rails at 0.80 in every winning fit and n_2 at 1.20 in the integrated and late-bin winners (1.12–1.19 elsewhere) — a two-sided calibration rail, stronger than in the earlier bursts, that travels as a caveat with every spectral statement below.

3. TIMING ANALYSIS

All timing runs on the approved selections, reference NaI n_5 .

3.1. Durations

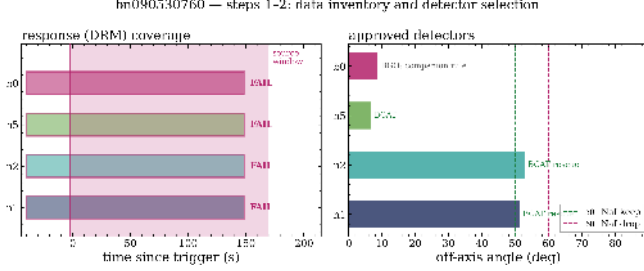


Figure 1. Detector selection for GRB 090530: source angles with the 50° rule; selected detectors n_1 , n_2 , n_5 , b_0 .

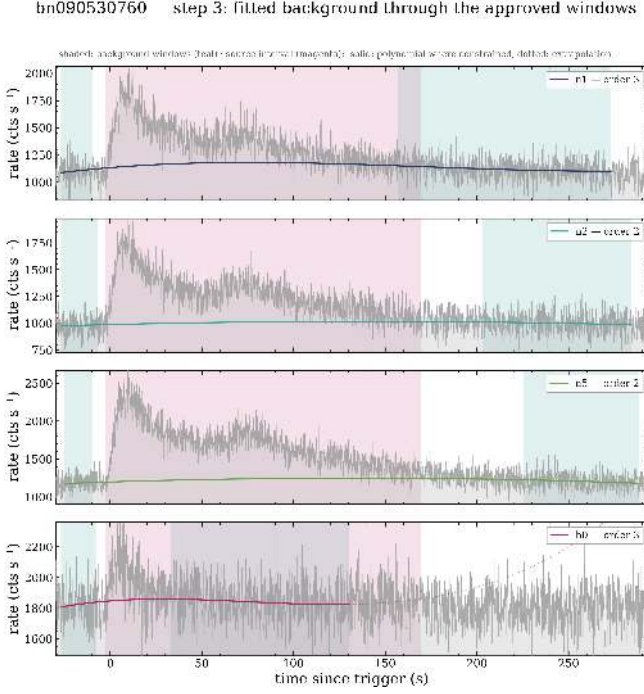


Figure 2. Approved background intervals and per-detector polynomial fits.

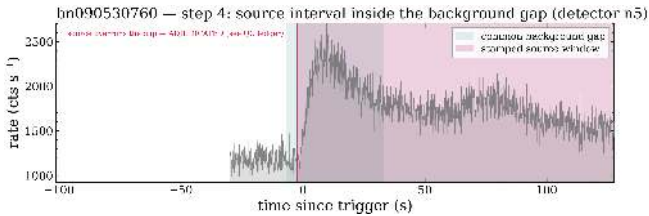


Figure 3. Approved source interval over the background-subtracted light curve.

The windowed duration is $T_{90} = 134.3^{+1.3}_{-0.7}$ s spanning $[5.13, 139.42]$ s, from the covariance-true Monte Carlo of the campaign estimator; the window is not truncated (the approved source interval extends to 169.2 s), so the duration is not a lower limit. $T_{50} = 68.9$ s. The T_{50}/T_{90} ratio of 0.51 already signals a decay-dominated profile, quantified next.

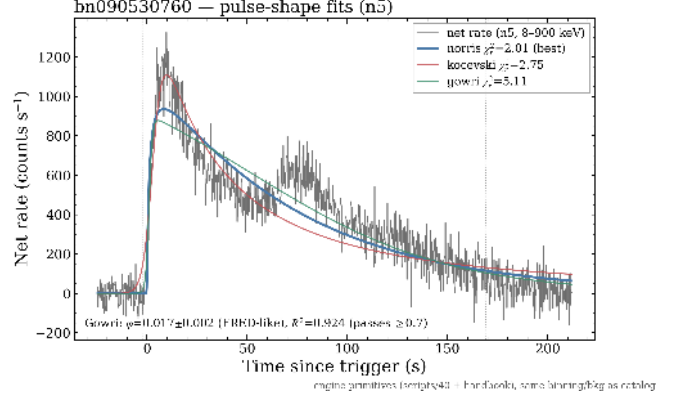


Figure 4. Pulse-shape fits to the net light curve; the Norris profile is the χ^2_r winner and the two-sigmoid asymmetry is $\varphi = 0.0169 \pm 0.0018$.

3.2. Pulse Shape

The Norris et al. (1996) profile wins the shape comparison for this burst, and the two-sigmoid of Gowri et al. (2025) gives $\varphi = s_l/s_r = 0.0169 \pm 0.0018$ ($R^2 = 0.92$): the most asymmetric FRED of the campaign so far, an order of magnitude beyond bursts #1 ($\varphi = 0.317$) and #2 ($\varphi = 0.263$) (Figure 4). The rise completes in a few seconds; the decay stretches over two minutes.

3.3. Minimum Variability Timescale

For this burst the three-estimator suite returns one bound and two honest failures (Figure 5): the Haar cross-check (Golkhou & Butler 2014) yields an upper limit < 8.0 s; the global continuous-wavelet method (Vianello et al. 2018) finds no significant minimum (reported as NaN, not as a value); and the windowed framework of Bala et al. (2026) refuses the burst at its admission gate (“no complete approval group in catalog”), a runner defect logged to the campaign register rather than silently skipped. A slow, smooth, two-minute FRED with no fast structure is exactly the regime where variability estimators should return limits; we treat the absence of a sub-second MVT as itself informative.

3.4. Spectral Lag

Scanning the cross-correlation fit window around the peak (half-widths 10.7–28.6 s; Figure 6) gives $\tau = 3.50^{+0.26}_{-0.25} \pm 0.48_{\text{win}}$ s (25–50 keV vs. 100–300 keV, soft lagging hard; detector n_5 , 16 ms resolution, peak significance 46σ), where the second uncertainty is the window-scan systematic. This is the largest lag of the campaign so far in absolute terms, yet as a fraction of duration ($\tau/T_{90} \approx 2.6\%$) it is *smaller* than burst #1’s 8.5% — a first hint that lag scales with pulse width sublinearly across the sample, one of the campaign’s accumulating cross-burst measurements.

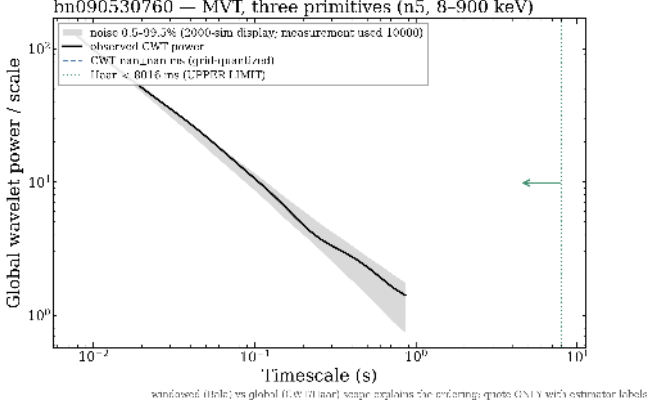


Figure 5. Minimum-variability estimators with explicit estimator labels; for this burst only the Haar upper limit (< 8.0 s) survives.

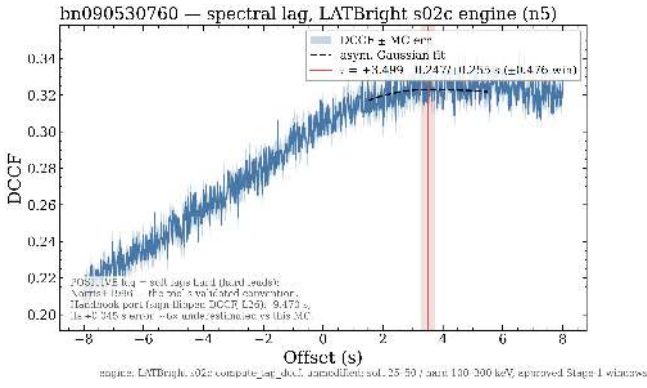


Figure 6. Cross-correlation lag with the window-scan systematic: members of the scan (half-widths 10.7, 21.5, 28.6 s) give $\tau = 4.10, 3.50, 3.15$ s; the quoted value is the median member and the half-spread is the ± 0.48 s systematic.

3.5. Per-band Durations and Energy Dependence

Per-band light curves and $T_{90}(E)$ are shown in Figure 7; the soft bands dominate the late decay, consistent with the large positive lag and the hard-to-soft spectral evolution of Section 5.3.

4. SPECTRAL BINNING

Six Bayesian blocks plus the time-integrated spectrum (Figure 8), from the frozen 3ML Bayesian-block + significance hybrid of the campaign (blocks computed 2026-08-12 and reused unchanged; re-deriving them per-convention would break cross-burst comparability).

5. TIME-RESOLVED AND TIME-INTEGRATED SPECTROSCOPY

5.1. Method

All 24 models of the campaign grid are fit to every bin (PGstat, AIC ranking (Akaike 1974)), with LLE/LAT bands absent for this burst (no high-energy data exist;

inclusion is gated on data availability, never on detection significance). The full 24×7 panel grid renders under the no-model-dropped rule with three provenance tiers; per-bin all-model parameter tables accompany the paper products. Winner panels for the key epochs are shown here; the complete montages are in the Appendix.

5.2. Integrated Spectrum: a Two-Break Candidate

The integrated spectrum (0.69, 111.86) s is won by the double-smoothly-broken power law of Ravasio et al. (2019) (Figure 9): $\alpha_1 = 1.96_{-0.04}^{+0.54}$, $E_b = 11.8_{-0.9}^{+1.1}$ keV, $\alpha_2 = -1.13_{-0.12}^{+0.08}$, $E_p = 69.9_{-2.7}^{+2.9}$ keV, $\beta = -3.06_{-0.14}^{+0.12}$ (model convention; AIC = 8659.4). Band follows at $\Delta\text{AIC} = 1.3$ — a statistical tie, so we report a two-break *candidate*, not a detection. The low-energy break at ~ 12 keV sits well above the 8 keV edge and the $E_b - E_p$ separation (factor ~ 6) matches the Ravasio-class morphology. This is the first burst of the campaign whose integrated spectrum prefers the survey’s namesake shape.

5.3. Time-Resolved Story: Violation, then Compliance

In the uniform Band convention the low-energy index runs $\alpha = +0.62 \rightarrow +0.47 \rightarrow +0.62 \rightarrow -0.05 \rightarrow -0.65 \rightarrow -0.54$ across blocks 0–5 while E_p decays $125 \rightarrow 36$ keV: the first three blocks sit *above* the synchrotron line of death ($\alpha > -2/3$), the last three fall to or below it. Neither of the first two campaign bursts showed this transition — burst #1 violated throughout, burst #2 complied throughout. Here the violation is a property of the rise and peak, not of the burst (Figure 10).

Model preference follows the same divide. The rise and peak prefer simple curved shapes (bin 1 Band, $\alpha = +0.46 \pm 0.09$, $E_p = 96.7$ keV; bins 2–3 SBPL); the decay prefers structure: bin 4 is won by DSBPL, narrowly (nearest competitor CPL+BB at $\Delta\text{AIC} = 0.8$, and the three nearest competitors are all BB-bearing — $E_b = 14.8_{-1.9}^{+4.4}$ keV, $E_p = 64.7_{-4.5}^{+5.2}$ keV; Figure 11), and in bin 5 the three top-ranked models each rail a slope bound (DSBPL α_1 at 2.5; DSBPLfree α_2 at 0.5; Band+BB α at 1.9) and are reported *invalid* under the campaign’s validity gate; the best valid model is CPL+BB ($kT = 4.5 \pm 0.4$ keV, $\Delta\text{AIC} = 4.1$ from the railed winner). We disclose the invalid ranking rather than silently re-ranking — the railed solutions are a bound-capped diagnosis (a known engine hardening item), and the bin-5 preference is reported as unresolved between a second spectral break and a cool thermal tail.

5.4. The Thermal Candidate

Block 0 (0.69, 4.94) s carries the campaign’s strongest photospheric candidate so far: CPL+BB with $kT =$

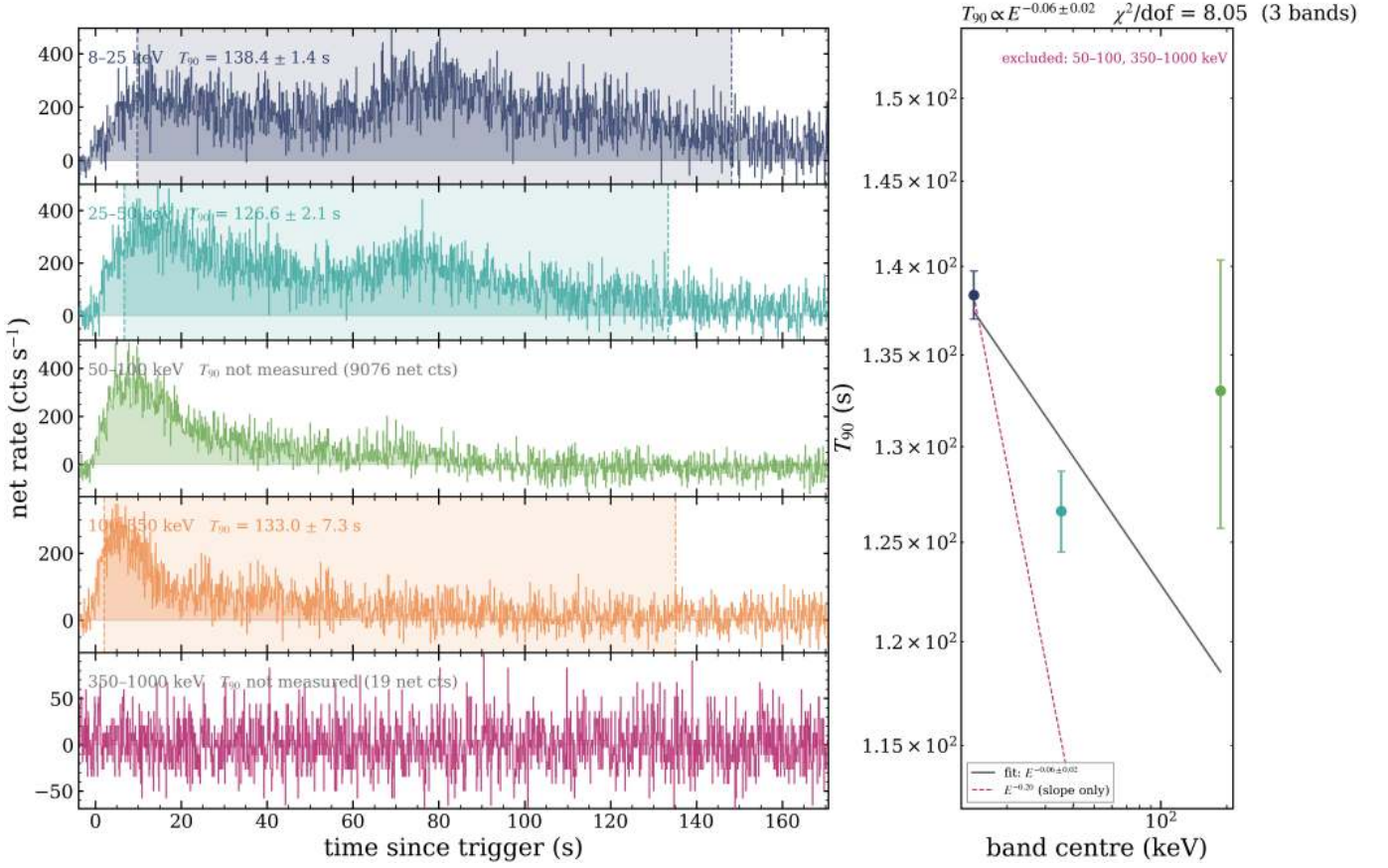
bn090530760 — step 7: energy-resolved T_{90} (detector n5); dashed = t_5, t_{95} 

Figure 7. Per-band light curves with individual T_{90} spans and the $T_{90}(E)$ relation; the population slope of Qin et al. (2013) is shown for reference only.

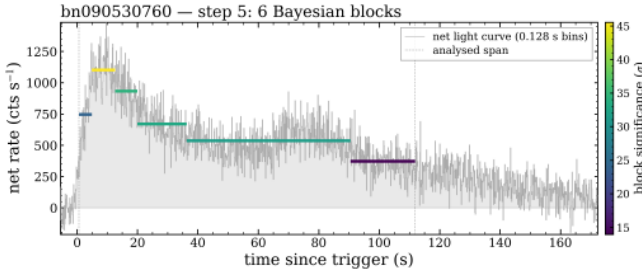


Figure 8. Bayesian-block spectral bins over the light curve.

22.4 ± 1.1 keV beats every BB-free model by $\Delta\text{AIC} = 3.1$ (best BB-free: DSBPL, itself bound-railed and flagged invalid; the best *valid* BB-free model trails by $\Delta\text{AIC} = 4.4$, strengthening the margin), and the second-ranked model is also BB-bearing (Band+BB, $\Delta\text{AIC} = 2.1$) — the thermal preference is model-robust (Figure 12). The Wien peak $3.92 kT = 88$ keV sits far from the 8 keV edge, so the edge-artifact failure mode of the campaign’s lesson list does not apply. With $\alpha = +0.62$ in the same epoch, block 0 is simultaneously line-of-death-violating

and thermally loaded — the photospheric configuration in which both signatures are expected together.

Two model-comparison metrics see the thermal component differently, and we report both. By AIC ranking, BB-bearing models win only blocks 0 and 5. By the nested likelihood-ratio gate of the step-9 scorecard (Figure 13), the blackbody is significant in blocks 0–4 (LRT = 43.9, 56.3, 32.0, 25.0, 11.2), with the temperature declining monotonically: $kT = 22.4 \rightarrow 20.4 \rightarrow 15.0 \rightarrow 12.1 \rightarrow 5.8$ keV — a cooling track spanning the burst, every Wien peak clear of the 8 keV edge. We report a tracked candidate with a cooling-photosphere morphology, pending the campaign-level residual-evidence test.

6. QUALITY CONTROL AND PROVENANCE

Step-9 QC (Figure 13) passes; every figure carries its producing script’s SHA-256 in a JSON sidecar; the numbers in this paper are recomputed from the product files, not transcribed from logs. Estimator failures (CWT, windowed-MVT admission) are reported as failures. All figures pass a fresh-context vision gate before delivery, enforced mechanically by a pre-delivery hook.

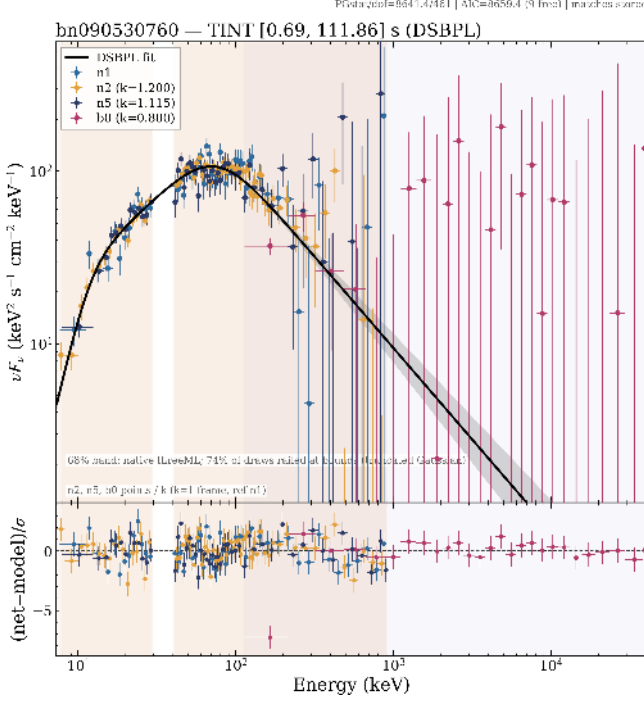


Figure 9. Integrated-spectrum winner: DSBPL with breaks at 11.8 and 69.9 keV. 68% error band from propagated posterior draws; PGstat/dof and the EAC rail are stamped on-figure.

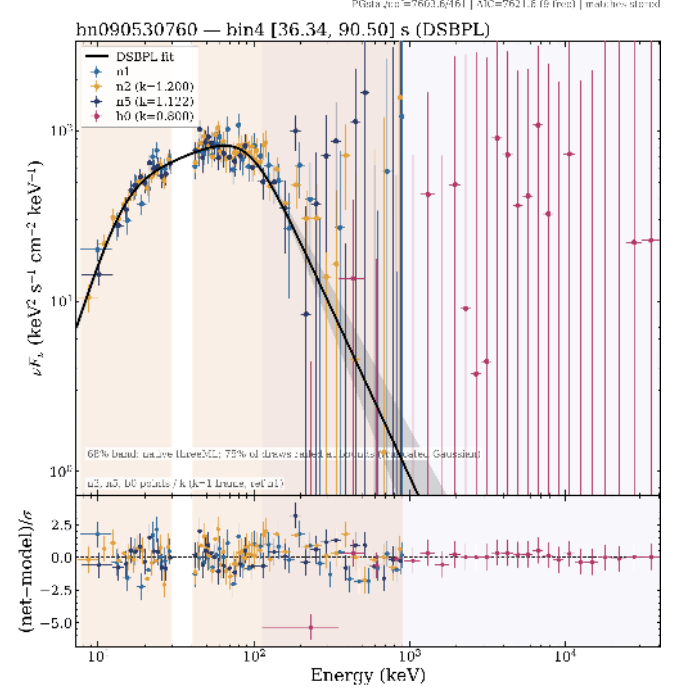


Figure 11. Block 4 (36.3, 90.5) s: DSBPL narrowly preferred (CPL+BB at $\Delta\text{AIC} = 0.8$) — the cleanest two-break epoch of the burst.

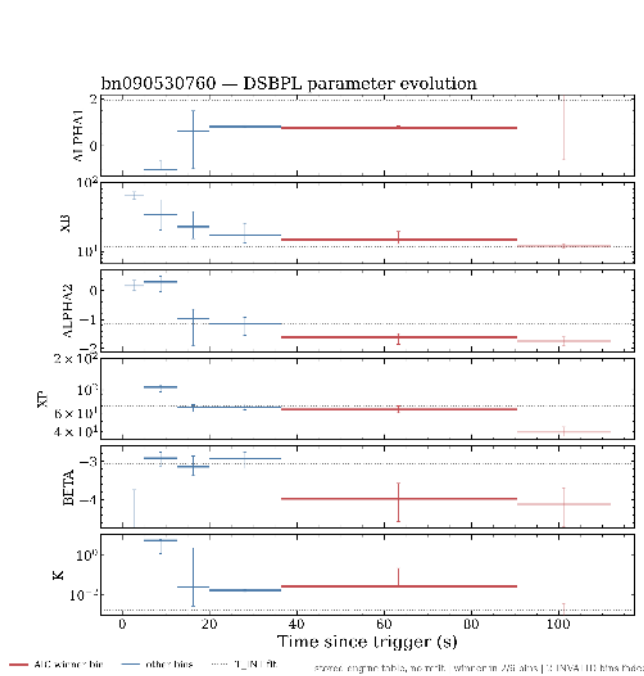


Figure 10. Parameter evolution: DSBPL family across the bins where it is constrained; the break energies and E_p decay monotonically.

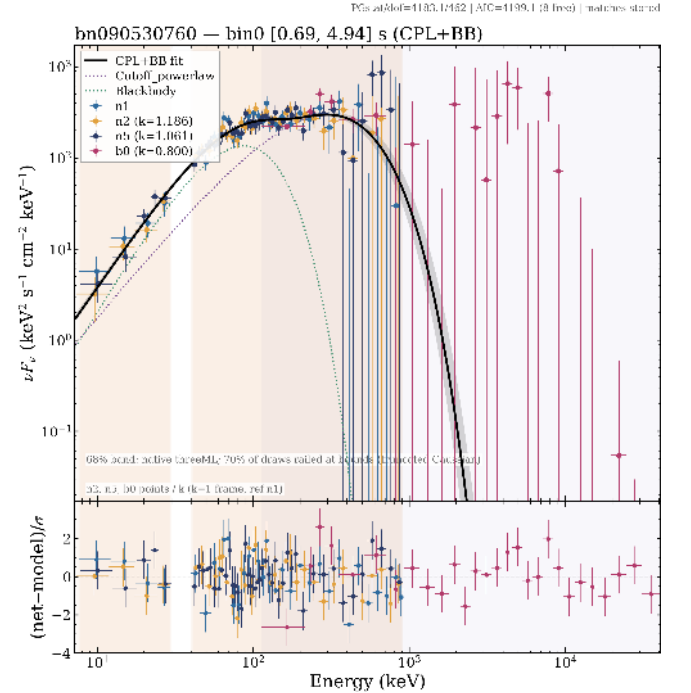


Figure 12. Block 0 winner CPL+BB: the $kT = 22.4$ keV blackbody (dotted components) carries the fit's preference over every BB-free competitor.

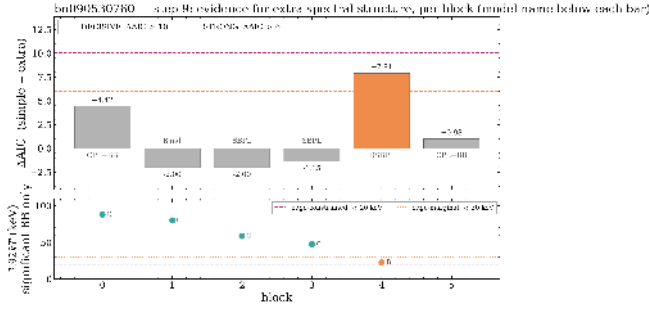


Figure 13. Step-9 scorecard: per-block evidence for extra spectral structure (top) and the 3.92 kT Wien-peak edge check for blocks with a significant blackbody (bottom) — the kT cooling track of Section 5.4 is visible directly.

Fits were produced by the campaign’s one-invocation 24-model engine run; the canonical table keeps the best optimizer minimum per (bin, model) across all available invocations.

7. SUMMARY

1. GRB 090530 is a two-minute, extreme-asymmetry FRED ($\varphi = 0.0169 \pm 0.0018$, the campaign’s most front-loaded pulse) with $T_{90} = 134.3^{+1.3}_{-0.7}$ s.
2. The spectral lag is the campaign’s largest in absolute terms ($\tau = 3.50^{+0.26}_{-0.25} \pm 0.48_{\text{win}}$ s) yet smaller than burst #1’s as a fraction of duration — lag does not scale linearly with pulse width.

3. No estimator resolves sub-second structure: Haar bounds the MVT from above (< 8.0 s) and the other two estimators return honest failures.
4. The synchrotron line of death is violated in the rise and peak (α up to $+0.62$) and respected in the decay — the violation is an epoch property, not a burst property.
5. Block 0 holds the campaign’s strongest thermal candidate ($kT = 22.4 \pm 1.1$ keV, $\Delta\text{AIC} = 3.1$ over all BB-free models, edge-gate clean); block 5’s best valid model is again thermal ($kT = 4.5$ keV) though its ranking is contaminated by bound-railed invalid fits, which we disclose.
6. The integrated spectrum is the campaign’s first DSBPL winner ($E_b = 11.8$ keV, $E_p = 69.9$ keV), a statistical tie with Band ($\Delta\text{AIC} = 1.3$): a two-break candidate in the [Ravasio et al. \(2019\)](#) morphology, with block 4 preferring the two-break shape outright.
7. Standing caveats: b_0 ’s effective-area factor rails at 0.80 in every winner and n_2 ’s at 1.20 in the integrated and late-bin winners; all numbers are provisional pending the PI’s campaign review.

APPENDIX

A. ALL-MODEL MONTAGES

Complete 24-model montages for the integrated spectrum and all six blocks (Figures 14–20); the winner panel is marked in each. Per-bin parameter tables for all models accompany the paper products.

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bn090530760 — TINT [0.69, 111.86] s — 24 models, AIC-ordered — winner DSBPL



Figure 14. Integrated spectrum: all 24 models.

bn090530760 — bin0 [0.69, 4.94] s — 24 models, AIC-ordered — winner CPLBB

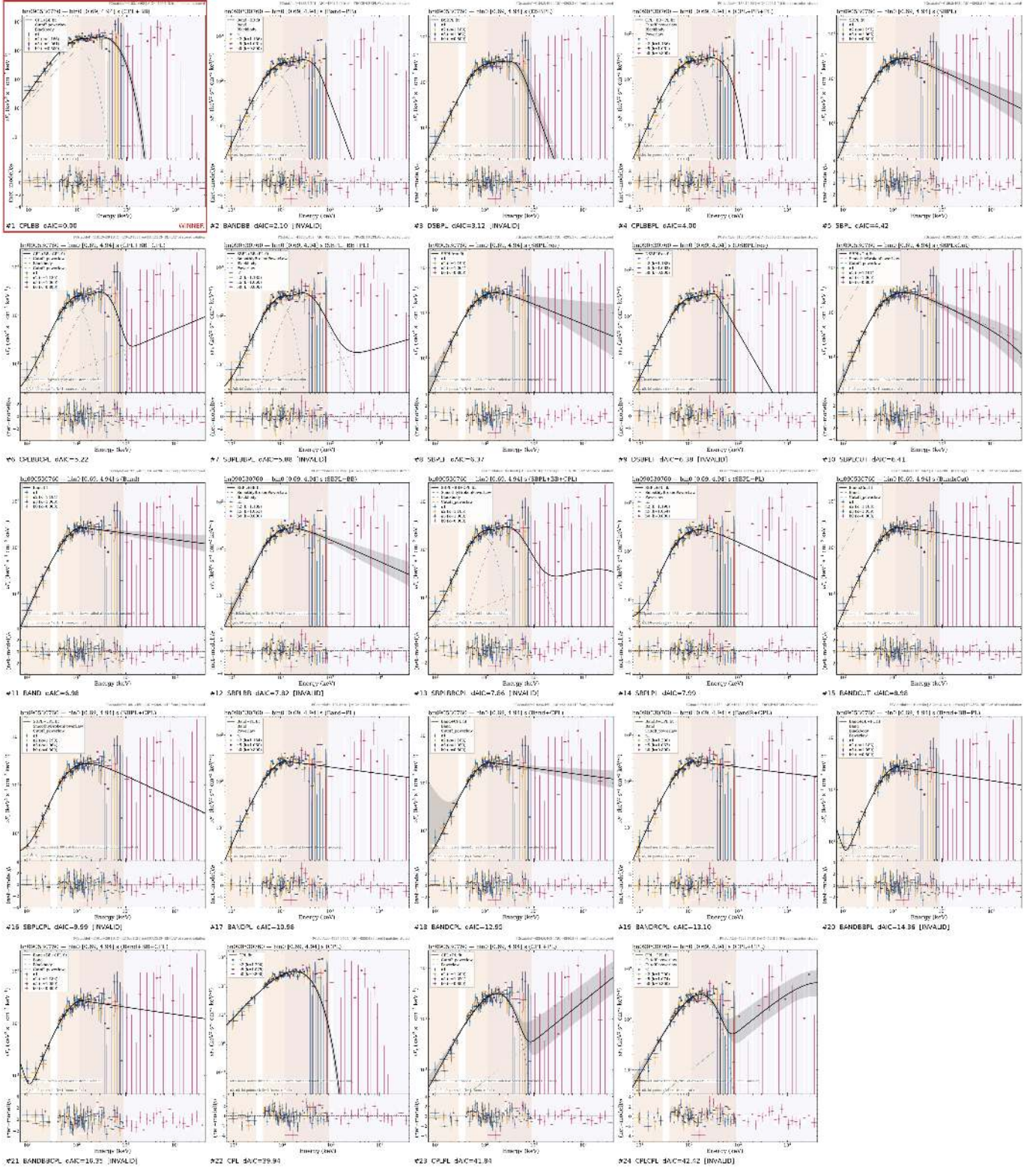


Figure 15. Block 0: all 24 models.

Figure 1 displays 25 panels showing the spectral energy distribution (SED) of the blazar 1ES 14.49+77.30. Each panel includes the following information:

- Top Panel:** Flux density (Jy) versus Energy (GeV) on a log-log scale. The plot shows data points with error bars and a model fit. The model parameters are listed in the top right corner of each panel.
- Bottom Panel:** Residuals (Data minus Model) versus Energy (GeV) on a log-linear scale.
- Model Name and AIC Value:** Each panel is labeled with a model name and its Akaike Information Criterion (AIC) value. The model names are: A1 BAND, A2 BANDCT, A3 SEPL, A4 CPLBE, A5 SEPLF, A6 BANDPL, A7 UNIDUC, A8 DBPL, A9 DBPLF, A10 DBPLCPL, A11 BANDCPL, A12 DBPLBR, A13 DBPLFPL, A14 DBPLCPL, A15 BANDCPL, A16 DBPLFPL, A17 DBPLCPL, A18 BANDEPL, A19 SEPLBEPL, A20 BANDESC, A21 SEPLBEPL, A22 CPLCPL, A23 CPL, and A24 CPL.
- Color Coding:** The panels are color-coded based on their AIC values. The 'WINNER' model (A1 BAND) is highlighted in red.

Figure 16. Block 1: all 24 models.

bn090530760 — bin2 [12.66, 20.02] s — 24 models, AIC-ordered — winner SBPL

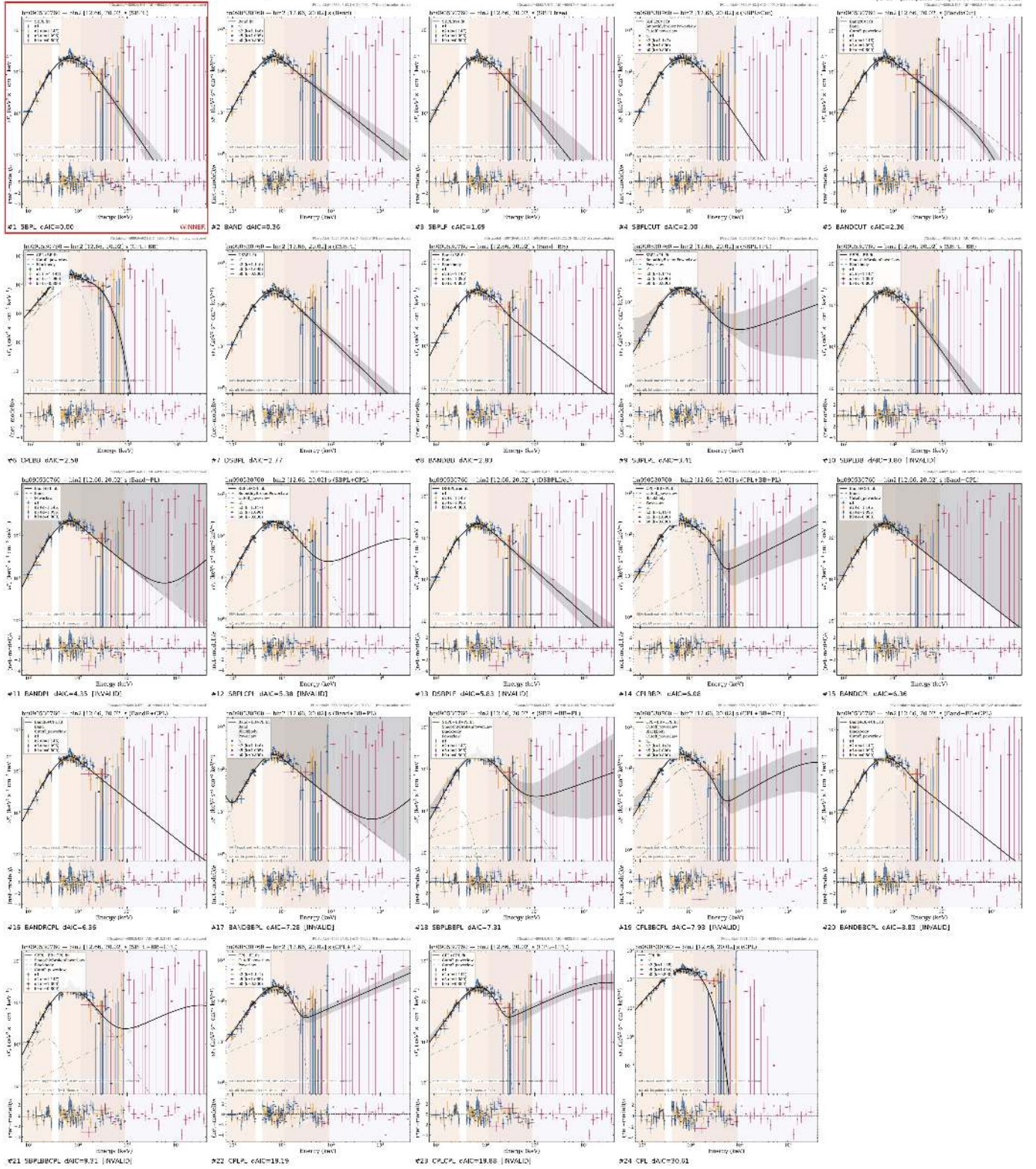


Figure 17. Block 2: all 24 models.

bn090530760 — bin3 [20.02, 36.34] s — 24 models, AIC-ordered — winner SBPL



Figure 18. Block 3: all 24 models.

Figure 1 displays 20 panels of spectral energy distribution (SED) plots, labeled #1 through #20. Each panel shows the flux density (Jy) versus energy (keV) on a logarithmic scale. The plots compare observed data points (black dots with error bars) with various model fits (solid black lines). The shaded regions represent the uncertainty or confidence intervals of the model fits. The x-axis ranges from approximately 10⁻² to 10² keV, and the y-axis ranges from 10⁻¹ to 10¹ Jy. The panels are arranged in a 4x5 grid. The first row contains plots #1 to #5, the second row #6 to #10, the third row #11 to #15, and the fourth row #16 to #20. Each plot includes a title indicating the model name and parameters, such as 'DSBP+GAC=3.00' for plot #1. The plots show a general trend of decreasing flux density with increasing energy, with some models showing a peak at lower energies and a tail extending to higher energies.

Figure 19. Block 4: all 24 models.

bn090530760 — bin5 [90.50, 111.86] s — 24 models, AIC-ordered — winner DSBPL



Figure 20. Block 5: all 24 models.